# A54- A11L.UAS.97: PROPOSE UAS RIGHT-OF-WAY RULES FOR UNMANNED AIRCRAFT SYSTEMS (UAS) OPERATIONS AND SAFETY

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16. Abstract :

The integration of Unmanned Aircraft Systems into the National Airspace System (NAS) necessitates a comprehensive reevaluation of right-of-way (ROW) rules. This research investigates the safety and operational implications of proposing ROW rules for UAS in low-altitude airspace, encompassing small UAS, UAS encountering other UAS, and swarms of UAS and crewed aircraft. The study employs a systematic approach, integrating literature reviews, gap analyses, simulations, and flight tests to derive data-driven recommendations. The findings highlight the applicability of the existing ROW rules of crewed aircraft to UAS operations and the necessity of updating traditional ROW rules, which have been historically grounded in the "see and be seen" principle and aircraft maneuverability, to accommodate the capabilities of detect-and-avoid (DAA) systems in ensuring safe separation and collision avoidance. The research underscores the significance of Beyond Visual Line of Sight (BVLOS) operations for the UAS industry. It identifies regulatory and research gaps that must be addressed to enable the safe application of ROW rules in BVLOS scenarios. Furthermore, the right-of-way rules for general encounter scenarios have been recommended systematically, along with data-driven maneuverability recommendations, rationale, and required standards. Additionally, the ROW influencers and aircraft handling characteristics have been discussed in detail. The research investigates the efficiency of remote ID to assist in avoidance maneuvers. The study proposes that non-ADS-B reservable airspace (NARA) be used to segregate airspace and mitigate conflicts between UAS and non-cooperative crewed aircraft. The research concludes by presenting recommendations for ROW rules in diverse UAS operational contexts, as well as suggestions for future research and regulatory development to foster the safe and efficient integration of UAS into the NAS.

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| Unmanned Aircraft Systems (UAS), Right-of-Way (ROW) Rules, Crash avoidance systems, Detect and Avoid (DAA), Beyond Visual Line of Sight (BVLOS), Reserved Airspace Concept (RAC), Non-ADS-B Reserved Airspace, Remote Identification (RID), Shielded operations. | This document is available to the U.S. public through the National Technical Information Service (NTIS), Springfield, Virginia 22161. This document is also available from the Federal Aviation Administration William J. Hughes Technical Center at actlibrary.tc.faa.gov. |

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## TABLE OF ACRONYMS

| Acronym | Meaning |
|---|---|
| A2C2 | Army Airspace Command and Control |
| AAM | Advanced Air Mobility |
| ADS-B | Automatic Dependent Surveillance-Broadcast |
| AGL | Above Ground Level |
| ARC | Aviation Rulemaking Committee |
| ASSURE | Alliance for System Safety of UAS Through Research Excellence |
| ASTM | ASTM International |
| BVLOS | Beyond Visual Line Of Sight |
| CFR | Code of Federal Regulations |
| DAA | Detect And Avoid |
| ERAU | Embry Riddle Aeronautical University |
| FAA | Federal Aviation Administration |
| FAR | Federal Aviation Regulations |
| GPS | Global Positioning System |
| KU | University of Kansas |
| LAANC | Low Altitude Authorization Notification Capability |
| NARA | Non-ADS-B Reserved Airspace |
| NAS | National Airspace System |
| NMAC | Near Mid-Air Collision |
| sNMAC | Near Mid-Air Collison of sUAV |
| NOTAM | Notice to Airmen |
| PIC | Pilot In Command |
| RPIC | Remote Pilot in Command |
| RAC | Reserved Airspace Concept |
| RID | Remote Identification |
| RoW | Right of Way |
| SPS | Standard Positioning Service |
| sUAS | Small UAS |
| UAM | Urban Air Mobility |
| UAS | Unmanned Aircraft System |
| UND | University of North Dakota |

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| UTM | Unmanned Aircraft System Traffic Management |
| WC | Well Clear |
| sWC | Small Well Clear |

## TABLE OF DEFINITIONS

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| Collision Avoidance | Collision avoidance involves preventing an intruder from penetrating a volume of airspace centered on the aircraft within which avoidance of a collision can only be considered a matter of chance (FAA, 2016; DoD, 2011). Collision avoidance is distinct from well clear in that well clear provides greater separation than collision avoidance. Collision avoidance can rely on both human and automated systems. The pilot uses proper scanning techniques, sounds (for Uncrewed Aircraft System (UAS) pilots), and vigilance. Automated systems include a sense and avoid system function where the Pilot in Command (PIC) is alerted to a conflict and manually takes action, or the UAS diverts to prevent a collision. |
| Cooperative intruders | Cooperative intruders carry equipment that allows the ownship to receive state information about the intruder. Electronic transmission of position information to include Mode C or Automatic Dependent Surveillance-Broadcast (ADS-B) are examples of cooperative technology. It is important to note that not all cooperative intruders are ADS-B equipped. ADS-B equipage is a subset of the larger set of cooperative aircraft. (Ramasamy, 2015) |
| Non-cooperatitve Intruders | Non-cooperative intruders are "silent," and all state data must be determined by sensors supporting the UAS operation, which may include both onboard and ground-based systems. (Ramasamy, 2015) |
| Detect and Avoid (DAA) | The capability of a UAS to remain well clear from and avoid collisions with other aircraft. (Federal Aviation Administration, 2009). |
| Mid-sized uncrewed aircraft | There is no standard definition of mid-sized uncrewed aircraft. However, for purposes of this research, a mid-sized UAS is one that is greater than 55 pounds but smaller than an aircraft capable of carrying a person. This can include aircraft such as the RMAX uncrewed helicopter, a ScanEagle, or the RQ-7 Shadow fixed-wing drone. The distinction for this research is not necessarily based on weight or size, however, but on conspicuity. |
| Remote Pilot in Command (RPIC) | The RPIC is a person who holds a remote pilot certificate with a small Unmanned Aircraft System (sUAS) rating and has the final authority and responsibility for the operation and safety of an sUAS operation conducted under part 107. |
| Reserved Airspace Concept (RAC)/Non-ADS-B Reserved Airspace (NARA) | A volume of airspace with defined boundaries and times within which particular rules apply and which particular aircraft might be operating within. This supports operations in controlled or uncontrolled airspace and conceptually exists as two types. First, a 3D polygon-shaped block of airspace; second, a 3D corridor defined by specified height, width, and length that can support beyond visual line of sight operations The intent the RAC or NARA is to segregate aircraft that cannot reasonably detect each other, specifically, to segregate crewed aircraft that are not equipped with ADS-B out, from uncrewed aircraft that cannot detect aircraft that are not equipped with ADS-B out. |

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| Right-of-way (RoW) (FAR 91.113) | The right of a vehicle to proceed with precedence over others in a particular situation. Right-of-way rules establish which aircraft in any encounter must give way to the other aircraft. 14 Code of Federal Regulations (CFR) § 91.113 is Right-of-way rules: Except for water operations. |
| Right-of-Way Violation (ROWV) | A right-of-way violation occurs when an aircraft, despite having the right of way, is compelled to change its course in order to avoid a collision with another aircraft. This implies that the other aircraft failed to yield as required, thereby infringing upon the right of the first aircraft to proceed on its intended path without obstruction. |
| See and Avoid (FAA-H-8083-3C) | See and avoid refers to the obligation conferred on each person operating an aircraft to maintain vigilance so as to see and avoid other aircraft. See and avoid includes the requirement to give way to aircraft with the RoW, and not pass over, under, or ahead of it unless well clear. 14 CFR Part B states that when weather conditions permit, regardless of whether an operation is conducted under instrument flight rules or visual flight rules, vigilance shall be maintained by each person operating an aircraft so as to see and avoid other aircraft. When a rule of this section gives another aircraft the RoW, the pilot shall give way to that aircraft and may not pass over, under, or ahead of it unless well clear. This concept relies on knowledge of the limitations of the human eye and the use of proper visual scanning techniques to help compensate for these limitations. Pilots should remain constantly alert to all traffic movement within their field of vision, as well as periodically scanning the entire visual field outside of their aircraft to ensure detection of conflicting traffic. A proposal in the Beyond Visual Line of Sight (BVLOS) Aviation Rulemaking Committee (ARC) Final Report (BVLOS ARC (FAA, 2022), 2022) recommends replacing this term with 'Detect And Avoid.' |
| See and Be Seen | Visual separation of air traffic depends on the principle of see and be seen, which requires that each person operating an aircraft maintain vigilance so as to see and avoid other aircraft and recommends that each person operating an aircraft make their own aircraft as visible as possible to other aircraft. The "See and Be Seen" concept incorporates both detection and conspicuity to enable safe interactions between aircraft. It is foundational to the principles of see and avoid, right-of-way, night lighting, and much of Part 91. The concept also underpins electronic detection and conspicuity systems, including transponders, TCAS, and ADS-B In/Out. |
| Sense and Avoid | Sense and Avoid is the capability of a UAS to remain well clear from and avoid collisions with other airborne traffic. Sense and avoid provides the functions of self-separation and collision avoidance to fulfill the regulatory requirement to see and avoid (DoD, 2011). |
| Shielded Operation | The FAA Drone Advisory Committee defines shielded operations as "flight within close proximity to existing obstacles and not to exceed the height of the obstacle" (Federal Aviation Administration, 2020c, pg. 31). Civil Aviation |

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| | Authority of New Zealand defines a shielded operation as one in which the "drone remains within 100 meters of, and below the top, of a natural or man-made object" (Civil Aviation Authority (CAA) of New Zealand, 2019). |
| Small Uncrewed Aircraft | Small Uncrewed Aircraft are small platform and associated elements (including communication links and components that controls the craft) that are required for the safe and efficient operation of such in the National Airspace System (NAS) (AIM, 2021). The actual aircraft must weigh less than 55 lbs. on takeoff including everything on board or otherwise attached (FAA, 2021). |
| Swarm | Swarms are biologically inspired collective robot systems, operate without centralized control, which uses local interactions with other robots and the environment as control inputs. Swarms use indirect communication from a leader robot to perform complex action or behavior. The disturbance to individual robots may not affect the overall ability or satisfy the collective goal (Leaf 2021). |
| Multi-Robot system | A multi-robot system consist of few agents which are assigned to do a specific task, which they cooperate to complete a goal. In a multi-robot system, each robot is able to do some sub-tasks of a given task. For such multi-robot system, it requires all the nodes (robots/drones) to reach the ultimate goal. |
| Well Clear (WC) | Well Clear is used in 14 CFR 91.113 to define the separation that a pilot must maintain between their aircraft and another aircraft with the RoW so as to not violate or interfere with the other aircraft's RoW. Part 91. states that when encounters occur, the aircraft that does not have the RoW shall give way to the aircraft with the RoW, and may not pass over, under, or ahead of the aircraft with the RoW unless well clear. The intent of Well Clear is to 1) maneuver in a manner that doesn't interfere with another aircraft's right of way and 2) maneuver in a manner that doesn't create a hazardous situation requiring collision avoidance. For DAA systems, a well clear violation also occurs when aircraft come closer than the DAA well clear separation criteria. A well clear violation is an indication that a right-of-way violation may have occured.A recommendation in the BVLOS ARC (FAA, 2022) proposes to replace this term with 'adequate separation.' |

# EXECUTIVE SUMMARY


The integration of Unmanned Aircraft Systems (UAS) into the National Airspace System (NAS) necessitates a comprehensive reevaluation of Right-of-Way (RoW) rules. This research investigates the safety and operational implications of proposing RoW rules for sUAS in low-altitude airspace, encompassing small Unmanned Aircraft System (sUAS), sUAS encountering other sUAS, swarms of sUAS, and crewed aircraft. The study employs a systematic approach, integrating literature reviews, gap analyses, simulations, and flight tests to derive data-driven recommendations. The findings highlight the applicability of the existing RoW rules of crewed aircraft to sUAS operations and the necessity of updating traditional RoW rules, which have been historically grounded in the "see and be seen" principle and aircraft maneuverability, to accommodate the capabilities of Detect-And-Avoid (DAA) systems in ensuring safe separation and collision avoidance.

The research underscores the significance of future Beyond Visual Line of Sight (BVLOS) operations for the UAS industry. It identifies regulatory and research gaps that must be addressed to enable the safe application of RoW rules in BVLOS scenarios. Furthermore, the right-of-way rules for general encounter scenarios have been recommended systematically, along with data-driven maneuverability recommendations, rationale, and required standards. Additionally, the RoW influencers and aircraft handling characteristics have been discussed in detail. The research investigates the potential use of Remote Identification (RID) technology for electronic conspicuity and detection.

The bulk of the simulations conducted were focused on the required detection distance capability needed to accommodate RoW rules. As such, the focus has been on defining a detection distance that reduces the severity of the encounter, as measured by the avoidance of a right-of-way violation. The determination of risk, then, requires a separate consideration of likelihood.

In one area, the team did not concur with the BVLOS Aviation Rulemaking Committee's (ARC's) initial recommendation to shift the legal burden from the sUAS to the crewed aircraft, requiring the crewed aircraft to give way to sUAS below 400ft Above Ground Level (AGL). The findings in this research address a temporary solution to alleviate the clear safety concerns related to this BVLOS ARC recommendation while still maintaining the position that sUAS should not have RoW below 400ft AGL.

A summary of the key findings in the report includes:

- Proposing Non-ADS-B Reservable Airspace (NARA) be used to segregate airspace and mitigate conflicts between sUAS and non-cooperative crewed aircraft below 400ft AGL.
- RoW safety hierarchy for RoW decision-making between sUAS and various sUAS and crewed aircraft with over 40 recommendations directly related to RoW rules in diverse sUAS operational contexts.
- RoW recommendations below 400ft AGL for interactions between sUAS, multiple-sUAS, and crewed aircraft. In part, recommendations include Head-On, Converging, Overtaking, and Emergency/distress encounters.
- An additional 11 recommendations for future research to include regulatory and standards development are needed to foster the safe and efficient integration of sUAS BVLOS below 400ft AGL into the NAS.

# 1 INTRODUCTION – RIGHT OF WAY REVIEW

All NAS users and operators have an equal, equitable, and shared legal responsibility (“duty”) to know and abide by the applicable and appropriate provisions of the Federal Aviation Regulations (FARs) and Aircrew Information Manual pertaining to their operation. Since 1998, the law in aviation is that those regulations, procedural, and informational documents are evidence of the standard of care among all pilots *(Management Activities Inc. v. US*, 21 F Supp 2d 1157, citing multiple, 1998.) and violations of FARs constitute negligence as a matter of law (*Id.*). The basic assumption in any right-of-way question, absent of special circumstances such as urgent or emergency status or categorical differences in aircraft type, is that all aircraft are assumed to be able to “see and avoid” others, no matter the situation.

That assumption and legal duty are contained in the existing right-of-way rules under 14 Code of Federal Regulations (CFR) 91. However, UASs are a new paradigm as they do not have an onboard pilot to see and avoid other aircraft, and in many cases, the UA is not visually conspicuous. Therefore, the existing legal duties of aircraft pilots to other aircraft do not work with the operations of aircraft to UAS. While the duty to see and avoid other aircraft remains unchanged, it must shift from the parties specified in current regulations to those capable of performing this function. This shift must be reflected in these recommendations.

Ordinarily, a legal duty rests affirmatively with the person acting or performing a task. When driving, for example, a person must act as a reasonable driver and drive safely, not speed, and avoid collisions. Other drivers are presumed to do the same, and the net result for everyone is increased safety. However, the law sometimes shifts legal burdens when special circumstances exist. For example, bicycle riders have a duty similar to that of car drivers to operate their bicycles safely. Some states shift the legal burden onto car drivers to take special care to avoid accidents with the more vulnerable bicyclists. That is one example of a burden shift. Another example of a burden shift—this time in aviation—is that professional flight crews are “charged with the knowledge which in the exercise of the highest degree of care they should have known” (*American Airlines v. US*, 418 F.2d 180, 193 (5th Cir 1969)), while private pilots have only a normal duty of care; i.e. they must act reasonably but might make mistakes. Another example to possibly consider is the shift in burden for Part 107 small drone operations to give way and remain well clear from crewed aircraft since crewed aircraft cannot effectively see-and-avoid small drones.

So, a burden shift of a greater duty onto someone who knows more or has greater technological ability is equally appropriate as shifting a duty away from someone who has less ability or capability. Both these principles apply in assessing the legal duties of UAS operators and other aircraft in the context of examining existing RoW rules and recommending changes to them.

An example of specific legal duties that have been codified into regulation in the RoW context is 14 CFR 91.113(f)-- Overtaking. An overtaking aircraft has the legal duty to avoid the overtaken aircraft by altering course to the right. While the direction of the maneuver is legally arbitrary and defined by regulation, the fundamental legal duty at issue, ordinarily shared equally by the pilots of both aircraft, shifts to the pilot of only one aircraft. That shift is based both upon the greater capacity of the overtaking craft to avoid a collision (i.e., greater information) and the context—in this case, relative positioning. Relative positioning in that the aircraft pilot being overtaken does not have an ability to visually see-and-avoid behind their ownship

In the aircraft overtaking example, both dissimilarities in information and physical location are key to the burden shift. Dissimilarities in information and physical location can also be used to differentiate legal burdens in a sUAS/sUAS or sUAS/aircraft situation and therefore inform appropriate right-of-way rules for new aircraft interactions not contemplated by existing regulations. For example, a sUAS that has DAA capability has different informational capabilities than one that does not, so the legal burden in that context could be different from that of a sUAS that lacks DAA capability. A DAA equipped sUAS and a non-DAA equipped sUAS looks the same to a crewed aircraft, even though they have very different senses and avoid capabilities. Therefore, it is appropriate to rely upon other attributes, such as position, when shifting the legal burden.

The right-of-way rules described in FAR 91.113 are based upon a shift in responsibilities between pilots in command of separate aircraft that may not be in active communication with one another. That shift occurs in response to (1) information dissimilarities (2) physical location, and 3) maneuver dissimilarity. FAR 91.113 standardizes those shifts by prescribing RoW rules between aircraft with varying abilities to avoid other aircraft. Adding sUAS to the existing RoW structure adds a new variable since they are small-uncrewed by definition. FAR Part 107.37 directs sUAS operators to yield the right of way to all other aircraft. Whether operated by a human or autonomous, sUAS must integrate with RoW in such a way that the existing burden shift regime remains predictable to non-communicating aircraft.

These principles applied to the data generated in this research have been used to provide RoW recommendations in consideration of UAS aircraft and emerging technologies.

# 2 RESEARCH QUESTIONS

During Task 2, the team identified initial recommendations for RoW rules. Three distinct areas were identified to conduct additional simulation and flight testing. These three areas or rounds were 1) Round 1 - General Interactions, 2) Round 2 - Reserved Airspace Concept, and 3) Round 3 - Remote ID. The investigation of these three areas created research questions that were related to, yet different and often more detailed in nature than the original research questions identified in the proposal. These additional questions and related use cases can be found in the simulation plan submitted as a deliverable for Task 3.

Furthermore, upon request by the research team, the FAA reviewed the initial research questions posed in the FAA's accepted research proposal and organized them by priority to the FAA.The FAA's reorganization of research questions assisted the research team's ongoing efforts to focus the work in a manner that maximized results and outcomes for the FAA. While all research questions were meaningful to the FAA, the organization of these research questions provided the degree of impact the questions were anticipated to have on current FAA activities (i.e., a 'tertiary research question' will still have an impact, just not as much of an impact as a 'primary research question').

The following sections provide that list, organized by Primary, Secondary, and Tertiary; Sub-bullets identify reference in either this document or in the FAA submitted "Preliminary/Draft and Interpretations Report for Task 3 and Task 4," herein referenced as "Task 3 and 4 Report" where the research questions were addressed.

## 2.1 Primary Research Questions

- What is the hierarchy of safety considerations, concepts, and requirements for establishing right-of-way rules for sUAS operations? What is the safety pedigree and the safety justification for right-of-way rules? Under what conditions are right-of-way rules needed? Under what conditions would additional right-of-way rules be unnecessary, burdensome, or overly prescriptive?
    - Hierarchy of Safety Considerations: Safety of Manned Aircraft, Safety of People and Property on the Ground, Safety of the UAS itself.
        - Reference: Final Report Section 2 and Section 9 - Introduction
    - Safety pedigree: The safety justification lies in the ability of the proposed rules to enable UAS to maintain safe separation distances from other aircraft, thereby mitigating the risk of mid-air collisions.
        - Reference: Task 3 and 4 Report, Section-4: Conclusion/Interpretation Of Simulation And Flight Testing Data; Final Report, Section 9
    - Under what conditions are right-of-way rules needed?
        - Reference: Task 3 and 4 Report, Section: 1; Final Report Section 1 and 9.
- What are the industry priorities for updating or creating new right-of-way rules to better integrate sUAS operations? What should regulators consider in the near-term vs in the longer term?
    - Reserved Airspace Concept (RAC) / Non-ADS-B Reserved Airspace (NARA)
        - Reference: Task 3 and 4 Report, Section 1.3 Reserved Airspace Concept (ROUND 2)
        - Reference: Task 3 and 4 Report, 3.2 Reserved Airspace Concept
        - Reference: Final Report, Section 8
- What are recommended right-of-way rules for unmanned aircraft that may be slightly less conspicuous than other manned aircraft in the airspace? What are the conspicuity metrics and thresholds for determining sUAS right-of-way? Consider the different DAA field of regard requirements for sUAS operating under Part 107.37 and UAS operating under Part 91.113 right-of-way rules. What data exists or can be measured to inform decisions?